STM32F407VET6 Board

U1.1
STM32F407VET6

ESP32-S3-DEVKITC-1-N16R8

U8
ESP32-S3-N16R8

R1 1kΩ
R2 1kΩ
R3 1kΩ
R4 1kΩ
R5 1kΩ
R6 1kΩ
R7 1kΩ
R8 1kΩ

HC-SR04

U7
HC-SR04

DHT22

U6
DHT22

BH1750FVI-TR

U3
BH1750FVI-TR

HX711

HX711_LOAD
1X4

HX711_MCU
1X4

MPU6050-TH

U4
MPU6050-TH

INA219AIDCNR

U5
INA219AIDCNR

Load Cell

U14
Load Cell
LOAD CELL
5 KG

ST-LINK_SWD

ST-LINK_SWD
1X4

F407_PWR
1X4

TITLE:	Sheet_1
REV:	1.0
Company:	Your Company
Sheet:	1/1
Date:	2026-10-21
Drawn By:	hafeeskhalid02

BH1750 module stand-in (C78960 symbol only): VCC=3V3, GND=GND, ADDR=GND (0x23), SCL=PB6, SDA=PB7, DVI=NC. Use a 1x5 header, not this footprint, for the PCB.

INA219 chip symbol (INA219AIDCNR) used only as a stand-in for the module. Real part: INA219 module with header. VS=3V3, GND=GND, SCL=PB6, SDA=PB7, A0=A1=GND (0x40), IN+/IN- to the load. Use a header footprint for the PCB.

DHT22 module stand-in (bare-sensor symbol): VCC=3V3, DATA=PC1 (pull-up on module), GND=GND, NC pin=no-connect. Check pin order on the real 3-pin module.

R2+R3 in series = 2k (Echo divider). R4-R8 in parallel = 200 ohm (INA219 test load)